

Ruibin Chen

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Brooklyn, NY

EDUCATION

New York University PhD in Electrical and Computer Engineering	Jan 2025 – present
New York University MS in Computer Engineering • Award: The Myron M. Rosenthal Award for Best MS Academic Achievement	Sep 2022 – May 2024 Brooklyn, NY, USA
University of Nottingham Ningbo China BSc Hons in Computer Science with Artificial Intelligence • First Class Honors • Honors: Head's Scholarship (2022)	Sep 2018 – July 2022 Ningbo, China

RESEARCH & PROJECT

Multi-Cell Multi-Band Handset Digital Twin for Learning-Based Closed-Loop Array Activation Research Assistant & First Author • Developed a Sionna ray-tracing-based UE digital twin combining NYU Tandon 3D urban geometry, multi-cell FR1/FR3 base stations, and an embodied eight-array handset with band-dependent antenna patterns. • Generated time-resolved CQI and rate traces under stochastic pedestrian mobility and handset self-rotation, capturing multi-band channel dynamics across feasible cell–array links. • Formulated an energy-constrained closed-loop array-activation problem and designed PPO-based rate-aware/risk-aware policies, achieving stronger best-rate performance than greedy and bandit baselines.	Sep 2026 – Present Brooklyn, NY, USA
UAV RCS simulation for Integrated Sensing and Communications Research Assistant • Implemented the UAV RCS channel simulation algorithm in NVIDIA Sionna and the UAV 3D random flying trajectory algorithm. Generating datasets for Integrated Sensing and Communications	Feb 2026 – Present Brooklyn, NY, USA
Transformer-Based Rate Prediction for Multi-Band Cellular Handsets Research Assistant & First Author • Developed a Sionna ray-tracing-based handset mobility simulator with a metal body model featuring dual FR3 (15 GHz) and FR1 (3.5 GHz) antennas, enabling UE translation and self-rotation on the NYU Tandon map to compute per-antenna channel capacity. • Formulated a multi-band channel rate prediction problem for frequency hopping under energy constraints, targeting accurate per-antenna rate prediction with sparse historical measurements. • Designed a transformer-based neural predictor combining temporal convolutional encoding and cross-antenna self-attention, achieving $\approx 28\%$ lower MSE and improved robustness under high-mobility fading for energy-efficient band selection.	Feb 2025 – Sep 2025 Brooklyn, NY, USA
TurtleBot-based FR3 TX localization Lead programmer • Managed the network configuration among the TurtleBot, visualization center, and FR3 receiver system, and bring up the FastDDS system for the ROS2 framework. Programmed the interface for turtlebot moving and PTUD48 Gimbal control. • Encapsulated the ROS2 topic publisher interface for the FR3 system to publish the Angle of Arrival to the ROS2 network. Design the slam visualization RViz add-on marker arrays to draw arrows for body orientation & AoAs, and the history lines to guess the Tx location at the line intersection points.	Oct 2025 – Present Brooklyn, NY, USA
Interpolation Techniques for Fast Channel Estimation in Ray Tracing Research Assistant & First Author • Proposed a reflection-model-based interpolation framework to accelerate ray-traced MIMO channel estimation. Implemented path clustering and kernel regression on reflected image sources to achieve high-fidelity spatial interpolation. • Conducted Sionna simulations in urban LOS/NLOS scenarios at 28 GHz, validating total received power and capacity accuracy across 1–8 m grids for wide aperture MIMO Transmitter Antenna Arrays.	March 2024 – Aug 2024 Brooklyn, NY, USA

Efficient and Plug-in Named Data Network (NDN) in Health Monitor System

Jan 2021 – Oct 2021

Research Assistant & Contribution Author

Ningbo, China

- Proposed the Neighborhood Registration Scheme to address producer mobility, reducing packet loss by over 80% and delay by 40% versus prior methods. Implemented and evaluated routing algorithms using ndnSIM (ns-3) and MATLAB, independently conducting experiments and analysis under faculty supervision.

Efficient Wireless Sensor Network Architecture Design

June 2020 – Aug 2020

Research Assistant & First Author

Ningbo, China

- Proposed an optimized clustering scheme by considering communication distance and residual energy. Selected weighted energy-efficient cluster heads to improve the overall lifetime of the Wireless Sensor Network. Completed simulation in MATLAB.

Breath Monitoring System Design

Nov 2022 – Dec 2022

Team Leader & Algorithm Engineer

New York City, USA

- Developed a hardware device for breath monitoring that detects asphyxia using CO₂ concentration and alerts users via LCD and LED indicators. Implemented an event-driven system on an STM32F4 platform with an SGP30 sensor using PlatformIO, and led a four-member team to successfully deliver the project.

Generation of 2.5D Word Cloud Algorithm

Sep 2021 – July 2022

Dissertation (Individual Final Year Project)

Ningbo, China

- Proposed a novel overlap detection algorithm and optimized 3D repulsive force model with minimal displacement; developed 3D visuals using OpenGL (C++), Cinema 4D, and Adobe Illustrator. Awarded First Class (Excellent).

Visualization and Validation Tool for Dynamic Models of Credit Risk

Sep 2020 – April 2021

Software Developer & Algorithm Engineer

Ningbo, China

- Developed visualization software that helped financial experts to validate the dynamic credit models, forecast loan default rates through different regression models, and assess credit risk under customized stress testing scenarios.
- Designed and operated the batch processing mode on SQL, improved data analysis efficiency by 40% through optimization of cloud and client data reading process. Utilized Java multi-thread framework to separate the UI thread and the data processing thread, which guaranteed the User experience.
- Learnt to analyze the requirement, generate specifications, communicate with the stakeholders, and design the personas.

PUBLICATION

- Chen, R., Lei, H., Guo, H., Mezzavilla, M., Poddar, H., Yoshimura, T., and Rangan, S., Transformer-Based Rate Prediction for Multi-Band Cellular Handsets, Accepted to IEEE ICC 2026 Workshop (WS02) — 4th Workshop on Intelligent Movable and Reconfigurable Antennas for Future Wireless Communication and Sensing. arXiv:2509.25722
- Chen, R., Joy, J., Hu, Y., Yin, M., Mezzavilla, M. and Rangan, S., 2024, October. Interpolation Techniques for Fast Channel Estimation in Ray Tracing. In 2024 58th Asilomar Conference on Signals, Systems, and Computers (pp. 1383-1388). IEEE.
- Kar, Pushpendu, Ruibin Chen, and Yukai Qian. "An efficient producer mobility management technique for real-time communication in NDN-based remote health monitoring systems." Smart Health 26 (2022): 100309.
- Chen, Ruibin, et al. "WLEACH-CK: Weighted K-means based LEACH-C algorithm for cluster head selection." 2021 17th International Conference on the Design of Reliable Communication Networks (DRCN). IEEE, 2021.

INTERNSHIP**China Telecom**

Oct 2024 - Dec 2024

Network Operations Intern

Shaoxing, China

- Analyzed cellular network performance using RSRP to assess coverage, handover behavior, and channel quality, supporting network optimization and stability improvements

SKILLS & INTERESTS

- Programming Skills: Python (7 yrs), C (4 yrs), C++ (4 yrs), MATLAB (4 yrs), Java, OpenGL, OpenCV, Git, JavaFX
- Simulation & Framework: NVIDIA Sionna, ROS2, Remcom Wireless Insite
- Robotics & Platform: Turtlebot4, PTUD48 Gimbal